

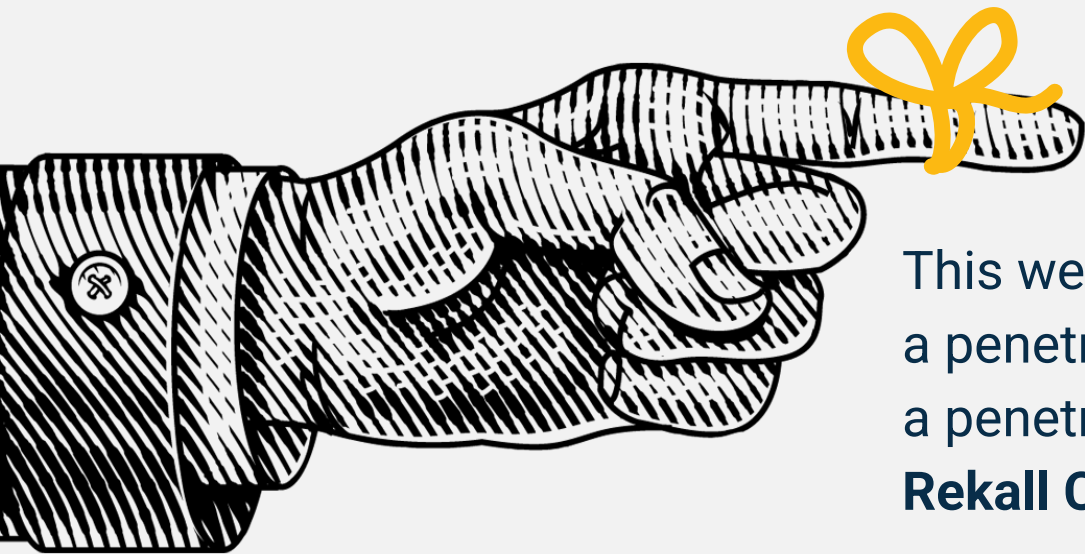
Cybersecurity

Project 2, Day 2: Linux Servers CTF

Lesson 17.2

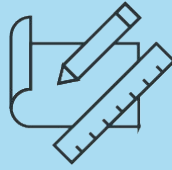


Welcome



Remember

This week, you're playing the role of a penetration tester hired to conduct a penetration testing engagement by **Rekall Corporation**.



Day 2 Objectives and Deliverables

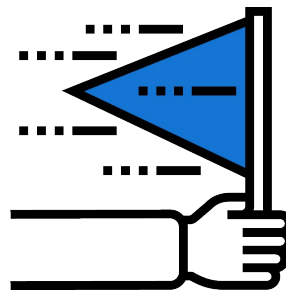


Project 2

On Day 1, you found vulnerabilities within Rekall's web application.

Previous Lesson

This Lesson



You'll continue finding "flags" during your exploitation.

Today, you'll continue exploiting Rekall's technical infrastructure, but you'll focus on Rekall's **Linux servers** and vulnerabilities related to Linux operating systems.



As you work through the project, you should continue completing the required deliverable: the **penetration testing summary report**.

Today's Lesson

Today's lesson will proceed as follows:





Important



While this is a group activity and you can share findings among group members, every student must submit this deliverable in order to receive credit for Project 2!



Day 2 Project Overview



Project Resources and Setup

CTF flag-submission page



You'll also access your CTF page within your Project 2 Lab.



You'll be provided with a custom website for your lesson's CTF when it's time to begin.



Once you have access to the page, register for an account.

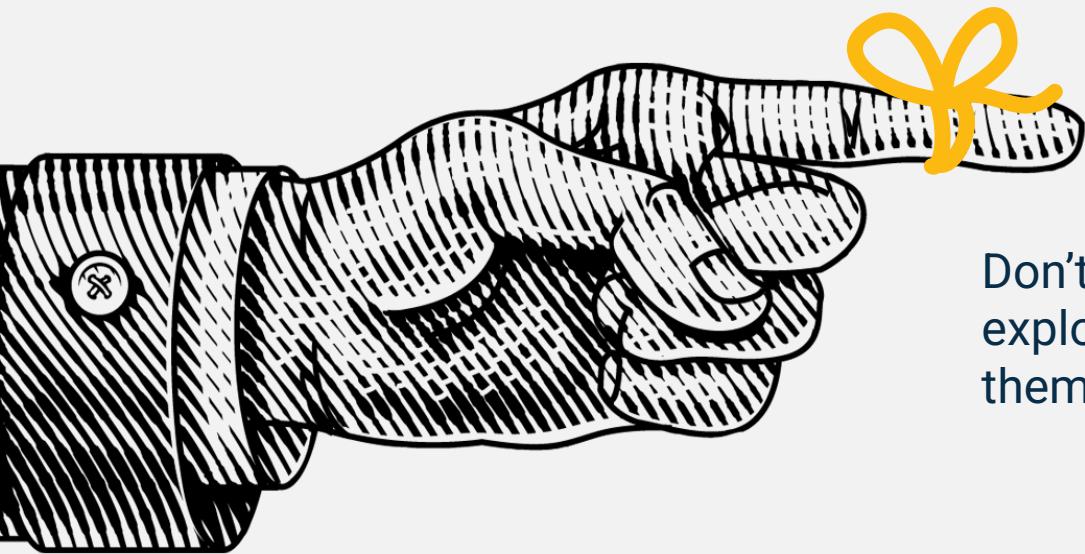


Remember your password in case you need to log back in!



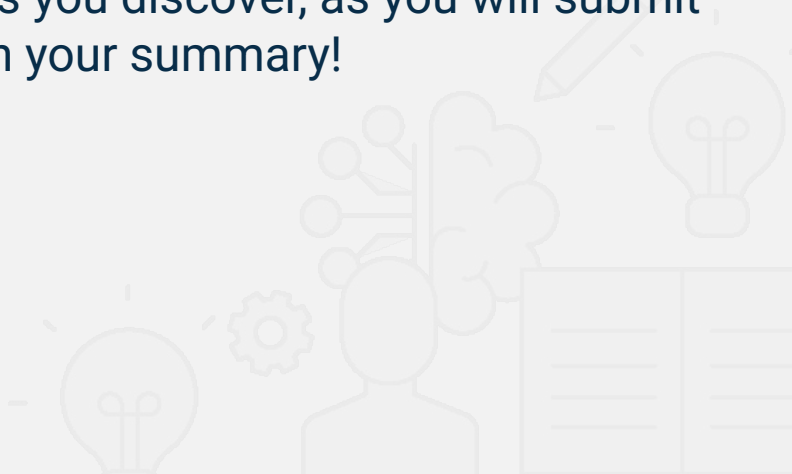
CTF Instructions

- The flags are hidden across various Linux hosts within Rekall's environment.
- Many flags are hidden within the Linux hosts and titled Flag 1, Flag 2, etc.
- Some flags are the answer to a specific challenge, such as: What is the IP of your machine?
- Each flag is composed of a string of letters and numbers, and is identified by a number (e.g., "Flag 1: d8sksydasksd").
- Once you find a flag, enter it on the CTF flag-submission page.



Remember

Don't forget to take **screenshots** of the exploits you discover, as you will submit them in your summary!





CTF Flag-Submission Page

This webpage contains the **12** available flags. Today's flag categories represent challenges associated with the phases of a penetration testing engagement.

For example:

Reconnaissance	This category contains challenges that use open-source intelligence tools.
Scanning	This category contains challenges that use scanning tools, such as Nmap and Nessus.
Exploitation	This category contains challenges that use exploitation tools, such as Metasploit.
Post-Exploitation	This category contains challenges that use post-exploitation tools, such as Meterpreter.

CTF Flag-Submission Page



The points awarded for each flag are indicated in its respective flag box. The more challenging the flag, the more points will be awarded.



Be sure to read the details in each flag for guidance on how to acquire that specific flag.



Once you find a flag, select the box with the corresponding flag number and enter the flag.



You can view your point total by selecting the Scoreboard option at the top of the page.

Expert	Flag 13 80			
	Flag Number Points Awarded			
Advanced	Flag 4 50	Flag 12 50	Flag 15 50	Flag 7 60
	Flag 14 60			



CTF Hints

This CTF provides you with an option to “pay” for hints with points.

Clicking a flag will also display any available hints for that flag and the points it costs to unlock those hints.



Note: You must already have been awarded the points before you can use them to “pay” for hints.

Challenge

0 Solves

×

Flag 13

80

PHP Injection

Unlock Hint for 30 points

Unlock Hint for 20 points

Flag

Submit

Hints for Success

-  Use the internet to help figure out methods to exploit these vulnerabilities.
-  Refer to your lesson notes and slides to help find several of the flags.
-  Certain exploits can be used to find information that will help you locate other flags.
-  Most of the exploits that you'll use were covered in class, but there are some new exploits that will require additional research.
-  Trying and failing is often part of a penetration tester's work, so don't be afraid to attempt **multiple exploits** until one is successful.

CTF Rules

Rules for today's activity:



Each group will start at the same time once the URL for the CTF has been provided.



The team with the most points at the end of the allotted time will win.



You can use all available resources to assist you: class notes, slides, internet resources, tools, etc.



If you are able to exploit a vulnerability and a flag does not display, please contact your instructor or TA to receive the flag.



Your instructor or TA can assist with technical or Lab access issues, but will not be able to provide guidance for finding the flags.



Questions?





Activity

Attacking Rekall's Linux servers

We are ready to begin the Linux server CTF!

Suggested Time
2 hours 20 minutes





Time's Up!
Let's review



Questions?





Next Class

Project 2: Windows CTF



Thank You